TRƯỜNG ĐẠI HỌC CÔNG NGHỆ ĐÔNG Á

VIỆN ĐÀO TẠO VÀ HỢP TÁC QUỐC TẾ

TIỂU LUẬN

HỌC PHẦN: QUẢN TRỊ DỰ ÁN CÔNG NGHỆ THÔNG TIN

MÃ ĐỀ THI: 04

TÊN ĐỀ TÀI: BE THE PROJECT MANAGER FOR: BUILDING A STUDENT INTERNSHIP MANAGEMENT AND PROGRESS TRACKING SYSTEM AT EAST ASIA UNIVERSITY OF TECHNOLOGY

LỚP TÍN CHỈ: QUẢN TRỊ DỰ ÁN CNTT-1-1-26(N01)

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PREFACE

In the current era of digital transformation, Information Technology has deeply integrated into every aspect of society, bringing breakthrough advancements in process automation and productivity enhancement. For higher education in general, and East Asia University of Technology in particular, the corporate internship is one of the most critical and transitional phases that allows students to gain practical experience. However, the management and progress tracking of internships are currently handled mostly manually, scattered across disjointed tools such as Emails, Excel spreadsheets, or hard-copy reports. This not only causes delays and overlaps in the operational processes of the Academic Affairs Office but also creates significant challenges for Instructors in tracking and evaluating individual student progress in real-time.

Stemming from this urgent need and a passion for optimizing business processes, I decided to choose the topic: "Building a Student Internship Management and Progress Tracking System (SITS) at East Asia University of Technology" for the final term paper of the IT Project Management course. The system is expected to become a centralized platform that seamlessly connects four key stakeholder groups: Administrators, Students, Academic Instructors, and Enterprise Mentors.

However, for a software project to evolve from a preliminary idea into a complete product, mastering programming skills alone is insufficient. The IT Project Management course has provided me with an entirely new perspective – the lens of a Project Manager. Through this term paper, I have had the opportunity to directly apply classroom theories to a practical project, ranging from establishing the Project Charter, developing the Work Breakdown Structure (WBS), and scheduling via AOA network diagrams and Gantt charts, to solving the complex problem of Resource Leveling and risk management.

I. PROJECT CHARTER

Project Name: Student Grade Management System (SGMS)
Project Code: PRJ-2026-ABC-SGMS
Date of Creation: August 19, 2026
Version: 1.0

Project Sponsor / Executive	Designated Project Manager
Name: Master Nguyen Thi Nga **Title:** Lecturer in the Department of Information Technology, EAUT **Email:** ngant2@eaut.edu.vn	**Name:** Le Duc Hai **Title:** Student, East Asia University of Technology **Email:** 20221526@eaut.edu.vn

1. BUSINESS CASE & PROJECT PURPOSE

Currently, internship management is handled manually and scattered across Email, hard copies, and Excel files. This leads to operational delays and makes it difficult for instructors to track student progress. Developing the Student Internship Tracking System (SITS) will modernize academic administration, ensure data integrity, and connect four user groups (Admin, Students, Instructors, Mentors) directly in real-time.

2. PROJECT OBJECTIVES & SUCCESS METRICS

Operational Efficiency: Complete 100% of the source code (Backend & Frontend) and successfully deploy it to the Cloud within exactly 12 weeks (60 working days). Documentation: Complete a graduation project report of 60-80 pages. System Reliability: The system operates stably with fast response times based on React.js, Node.js (Express), and relational database technologies.

3. HIGH-LEVEL SCOPE

3.1. In-Scope Account management and Role-Based Access Control (RBAC) using JWT. Internship registration process; weekly report submission and approval. Performance evaluation and grading; data statistics and report export (Excel/PDF).

3.2. Out-of-Scope Do not develop real-time messaging features. Do not process financial transactions or tuition fees.

4. HIGH-LEVEL MILESTONES & TIMELINE

Milestone	Key Milestone Activity	Target Date
M1	Initiation and Requirement Elicitation	Week 1 - 3
M2	System Analysis and Design	Week 4 - 6
M3	System Development	Week 6 - 9
M4	Testing and Optimization	Week 9 - 10
M5	Production Deployment & Documentation	Week 11 - 12

5. BUDGET & RESOURCE ALLOCATION

Total Approved Budget: 0 VND (Student self-funded project) / Cloud Server rental costs are covered by the team. Funding Source: Contributions from the student project team. Assigned Key Resources: Project Manager (Le Duc Hai) - 100% allocation Backend Developer (Member 2) - 100% allocation Frontend / QA Developer (Member 3) - 100% allocation

6. PROJECT MANAGER AUTHORITY LEVEL

The designated Project Manager (Le Duc Hai) is granted authority by the Project Sponsor to: Coordinate internal resources and assign project-related tasks. Approve schedule changes within the permitted threshold. Ensure the project complies with technical requirements and the 60-day timeline.

7. KEY RISKS & MITIGATION STRATEGIES

Key Risk	Severity	Mitigation Strategy
API integration conflict between Frontend and Backend	High	Unify API documentation standards (Swagger/Postman) from the beginning and perform continuous code merging
System authorization security flaw (RBAC, JWT)	Medium	Thoroughly test authorization Middlewares for each user group before UAT.
Schedule delay due to inaccurate coding time estimation	High	Conduct daily Stand-up meetings to closely monitor the workflow progress on the critical path

8. STAKEHOLDERS & GOVERNANCE

Project Steering Committee: Course Instructor (MSc. Nguyen Thi Nga). Primary End Users: Admin (Academic Affairs), Students, Instructors, Industry Mentors. Reporting Governance: Weekly regular progress reports to the Course Instructor; daily internal team meetings.

9. AUTHORIZATION & SIGN-OFF

By signing below, the Project Sponsor authorizes the official initiation of the Student Internship Tracking System (SITS) project and empowers the Project Manager to apply organizational resources toward its execution.

Project Sponsor:

_____

Name: Master Nguyen Thi Nga

Title: Lecturer in the Department of Information Technology, EAUT

Date: _____

Project Manager:

_____

Name: Le Duc Hai

Title: Student, East Asia University of Technology

Date: _____

II: PROJECT SCHEDULE PLANNING

1. Work Breakdown Structure (WBS) & Activity List

#	WBS Item	Complexity	Duration (days)	corresponding name
1.	**Initiation and Requirement Elicitation**	Medium	**15**	**A**
1.1	Survey current internship management status	Simple	4	A1
1.2	Gather & define system requirements (SRS)	Medium	6	A2
1.3	Draft project outline & WBS	Simple	5	A3
2.	**System Analysis and Design**	**Medium**	**12**	**B**
2.1	Analyze functions & business workflows	Simple	4	B1
2.2	Database design (6 core tables)	Medium	4	B2

#	WBS Item	Complexity	Duration (days)	corresponding name
2.3	RESTful API architecture design	Simple	4	B3
2.4	UI/UX design	Medium	4	B4
3.	**System Development**	**Medium**	**16**	**C**
3.1	Build Backend module (Node.js)	Medium	14	C1
3.2	Build Frontend module (React.js)	Medium	16	C2
4.	**Testing and Optimization**	**Medium**	7	**D**
4.1	Functional testing (4 modules)	Simple	3	D1
4.2	Security testing (RBAC, JWT)	Medium	2	D2
4.3	Bug fixing & Database optimization	Medium	2	D3
5.	**Deployment and Documentation**	**Medium**	**10**	**E**
5.1	Deploy system to	Medium	3	E1

#	**WBS Item**	**Complexity**	**Duration (days)**	**corresponding name**
	Cloud			
5.2	Draft full thesis report	Medium	7	E2
5.3	Prepare presentation slides & Demo video	Simple	3	E3

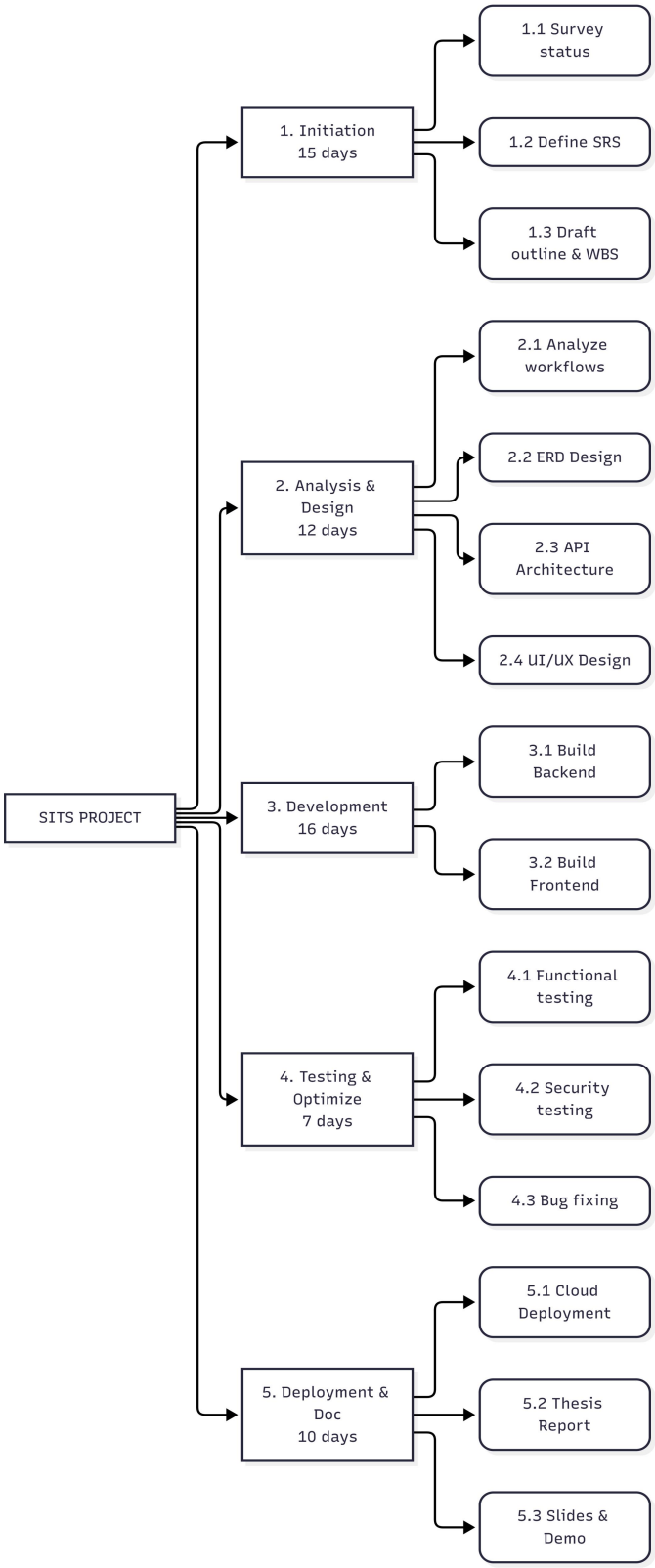


Image 1: Work Breakdown Structure (WBS)

2. Network Diagram (AOA) & Critical Path

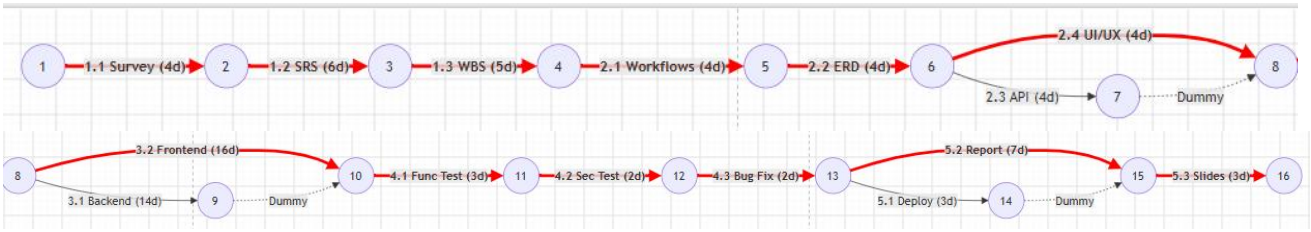


Image 2: Network Diagram (AOA)

Flow analysis: In Phase 2, tasks 2.3 and 2.4 run in parallel. In Phase 3, tasks 3.1 (12 days) and 3.2 (13 days) run in parallel. In Phase 5, tasks 5.1 (2 days) and 5.2 (6 days) run in parallel.

Critical Path: Based on the AOA method, the longest sequence of tasks determining the project schedule passes through the following nodes: 1.1 ➔ 1.2 ➔ 1.3 ➔ 2.1 ➔ 2.2 ➔ (2.4 or 2.3) ➔ 3.2 (Frontend) ➔ 4.1 ➔ 4.2 ➔ 4.3 ➔ 5.2 ➔ 5.3.

Completion time: The total project duration is 48 working days. The Backend task (3.1) has a float of 1 day; the Cloud Deployment task (5.1) has a float of 6 days.

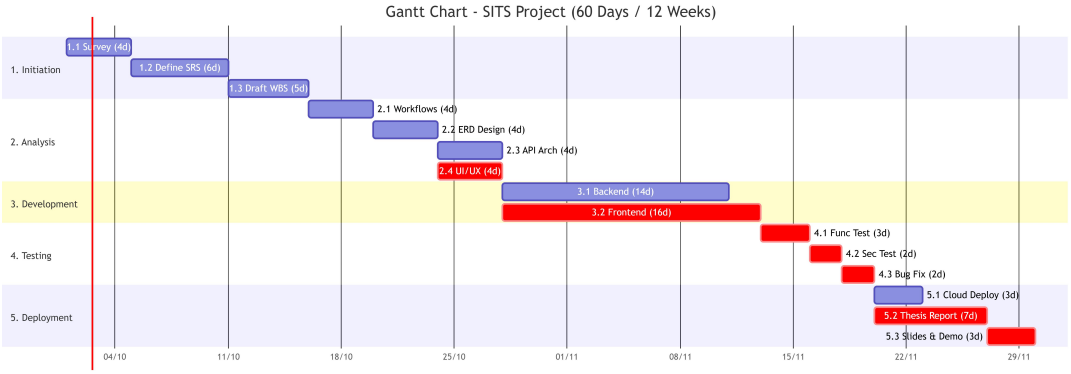


Image 3: Gantt Chart

III: RESOURCE & HUMAN RESOURCE MANAGEMENT

1. Work Allocation & RACI Matrix

Deliverable / Task	**Member 1 (PM)**	**Member 2 (Backend)**	**Member 3 (Frontend / QA)**
Collect requirements (SRS)	A / R	C	C
Business Analysis (ERD & API Design)	A	R	C
UI/UX Analysis & Design	A	I	R
Code function (Backend API)	A	R	I
Code function (Frontend Web SPA)	A	I	R
Test (Functional & Security Testing)	A	C	R
Reporting & System Handover	A / R	R	C

2. DISC Analysis and Team Management Strategy

Member 1 (PM) - Group D (Dominance): Focused on meeting the 48-day deadline; decisive. Strategy: Delegate overall coordination and project scope finalization; closely monitor the Critical Path; limit the PM's direct involvement in detailed coding.

Member 2 (Backend) - Group C (Compliance): Logical thinker; prefers working independently with data; adheres to database design principles. Strategy: Assign tasks related to ACID-compliant architecture design and JWT security. Require precise API documentation (Postman docs) to serve as a reference point during code integration.

Member 3 (Frontend/QA) - Group I (Influence): Creative; UI-oriented; enthusiastic. Strategy: Assign UI/UX design tasks and QA responsibilities (creating test cases). Encourage frequent presentations and product demos.

3.Workload Histogram & Resource Leveling

To optimize performance, the team analyzes resource loads (measured in required personnel) during critical project milestones and subsequently develops a leveling strategy to prevent personnel overload.

Data Table 1: BEFORE LEVELING

Timeline (Days)	**PM Required (Person)**	**Backend Required (Person)**
Day 01 - 15	1	0
Day 16 - 27	1	1
Day 28 - 43	1	1
Day 44 - 50	1	1
Day 51 - 53 (Overallocated)	2	0
Day 54 - 60	1	0



Image 4: Before Leveling

Data Table 2: AFTER LEVELING

Timeline (Days)	**PM Required (Person)**	**Backend Required (Person)**
Day 01 - 15	1	0
Day 16 - 27	1	1
Day 28 - 43	1	1
Day 44 - 50	1	1
Day 51 - 53 (Leveled)	1	1
Day 54 - 60	1	0

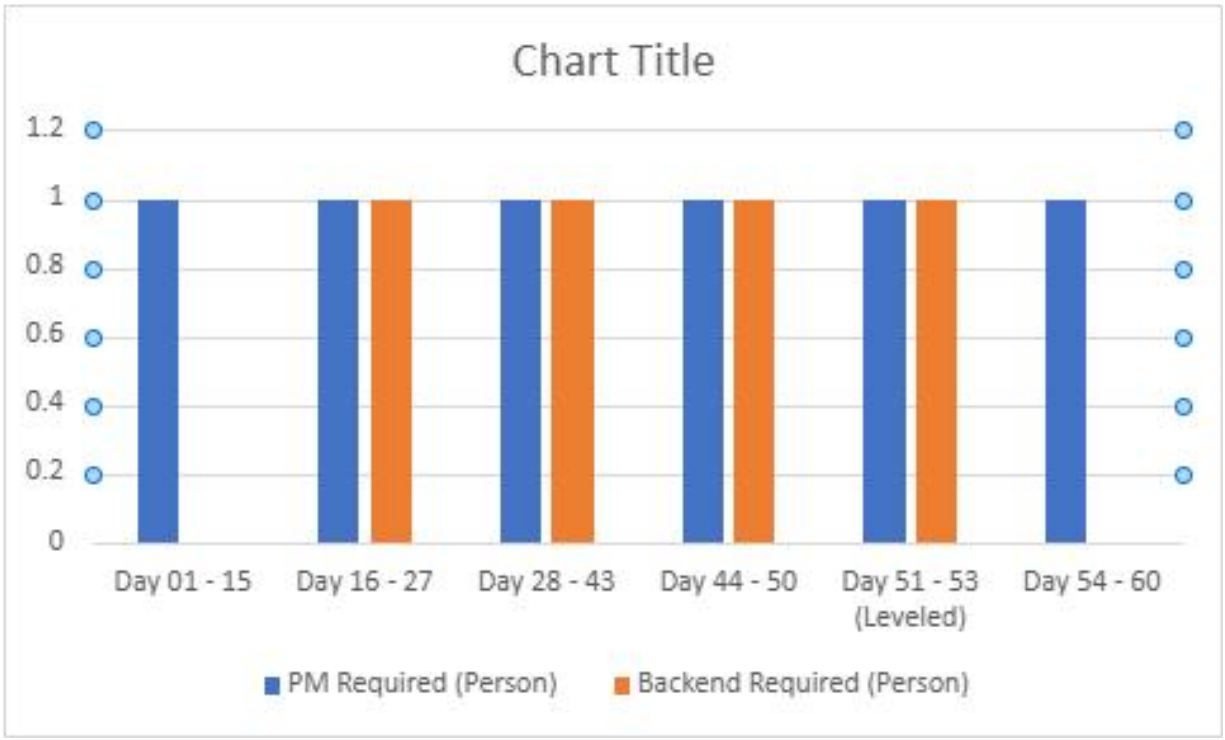


Image 5: After Leveling

Resource Leveling Strategy:

Based on Data Table 1, between Day 51 and Day 53, the project requires 2 Project Managers to simultaneously execute Task 5.1 (Deploy Cloud) and Task 5.2 (Thesis Report). Since the team only has 1 PM, this results in a direct overallocation. Meanwhile, the Backend developer is completely idle (0 personnel required).

By applying the resource leveling technique based on Float prioritization, the PM reallocated Task 5.1 (which has a 4-day Float and is not on the critical path) to the Backend developer. The PM now focuses solely on Task 5.2 (which is on the Critical Path). The result in Table 2 shows that the workload has been perfectly balanced to require exactly 1 PM and 1 Backend developer, fully resolving the overload risk while strictly maintaining the 60-day project schedule.

IV: RISK MANAGEMENT & PROJECT CLOSURE

1. Distinguishing between Risks and Issues

Risk: An event that has not yet occurred but has a certain probability of happening and would impact the project (either negatively or positively). (Example: Potential code conflicts during Frontend and Backend integration).
Issue: A risk that has materialized; it is currently having a direct impact and requires immediate attention from the Project Manager. (Example: Cloud server failure on the day of the demo).

2. Risk Register

Identify key risks and propose response plans, including estimated time and costs for mitigation or resolution.

Risk Item	Risk Level	Prevention	Recovery	Responsible Person	Time Cost
API synchronization mismatch	High	Write Swagger/Postman docs and unify Data Formats early	Hold emergency stand-up meeting to finalize API Output standards	PM	4 hours
Source code merge conflict	Medium	Define clear branching rules on GitHub	Temporarily stop pushing code, review line-by-line to Resolve Conflict	T2 (Backend)	2 hours

Risk Item	Risk Level	Prevention	Recovery	Responsible Person	Time Cost
Unexpected team member absence / illness	High	Push code to GitHub daily, apply Cross-training	Reallocate tasks to remaining 2 members, use buffer time	PM	16 hours
RBAC authorization security flaw	Low	Thoroughly test Role checking Middleware and JWT Tokens on each API	Temporarily disable flawed API flow, release hotfix	T2 (Backend)	3 hours
Poorly written testing documentation	Medium	Create detailed Test Cases for 4 modules immediately after MVP code	Present the most thoroughly tested scenarios, note feedback	T3 (QA)	5 hours

3. Project Closure

Product Acceptance & Handover:

Hand over backend and frontend source code and the database to a GitHub repository, including a `README.md` file with local installation instructions.

Submit the project report (Word/PDF, 60–80 pages) and the test case matrix for review and acceptance.

Conduct an in-person defense before the review panel using PowerPoint slides and a functional demo video (5–7 minutes).

Team Performance Evaluation: Hold an internal retrospective meeting where the three members conduct peer reviews based on code contribution volume and adherence to deadlines.

Lessons Learned:

Working in a three-person team requires communication documentation (API docs, DB schema) to be highly accurate and finalized at the outset to prevent the frontend team from having to wait for the backend team to modify APIs.